

Rafael Reis

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EDUCATION

McGill University

Master of Engineering in Applied Artificial Intelligence

Montreal, QC

Aug. 2026 – Exp. May 2028

McGill University

Bachelor of Engineering in Computer Engineering, Minor in Applied Artificial Intelligence

Montreal, QC

Sep. 2021 – May 2026

EXPERIENCE

Undergraduate Research Assistant for Dr. Derek Ruths

Jan 2026 – May 2026

Network Dynamics Lab / McGill University

Montreal, QC

- Built failure-mode-aware counterspeech generation pipelines using GPT-4.1-mini via OpenAI's API, mitigating recurring failure patterns to improve response safety
- Built a structured evaluation framework benchmarking baseline vs. fine-tuned LLMs to quantify quality tradeoffs across generation strategies
- Processed large-scale hate speech datasets in Python, engineering prompts and metrics to compare model outputs at scale

Undergraduate Research Assistant for Dr. Amir Shmuel

May 2025 – Aug 2025

The Neuro / McGill University

Montreal, QC

- Optimized MATLAB/Python data pipelines for Optical Imaging, reducing analysis time and improving reproducibility
- Developed scripts to extract and quantify neural activity from Optical Imaging and fMRI datasets for lab research
- Designed and 3D-printed custom imaging fixtures in Blender and UltiMaker Cura to support functional imaging experiments

DevSecOps & Cloud Security Intern

May 2024 – Aug 2024

Canadian National Railway / CISO

Montreal, QC

- Designed and deployed CI/CD pipelines in Azure Pipelines to automate build, test, and deployment for backend security systems using Python, Terraform, YAML, and PowerShell
- Wrote Terraform IaC to provision and maintain cloud resources, reducing manual configuration errors across environments
- Hardened code infrastructure supporting CN's enterprise security network, reducing attack surface

Server & Bartender

May 2026 – Present

Breizh Cafe

Montreal, QC

- Independently ran front-of-house for a 41-seat restaurant, averaging 50 covers/weekday shift solo and 90/weekend shift, handling opening/closing and cash flow
- Delivered full bar and table service under high-volume, time-pressured conditions, earning an average 17.6% tip rate

Academic Tutor

2018 – Present

McGill University & Self-Employed

Montreal & Paris

- Provided academic support to university-level students at McGill, tutoring subjects such as Calculus II, Linear Algebra, Differential Equations and intro to Python
- Enhanced teaching and communication skills by delivering weekly lessons to a network of middle and high school students in France

IT Risk Management Intern

May 2022 – June 2022

Groupe BCPE & Natixis

Paris, France

- Tracked budgets in Excel and Python across two large-scale IT projects run by five branches, supporting resource allocation and future planning

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C#, JavaScript, SQL, HTML/CSS, ARM Assembly, VHDL
Frameworks/Libraries: React, Unity, PyTorch, TensorFlow, NumPy, Pandas, Scikit-Learn, HuggingFace
Developer Tools: Git, Docker, Terraform, YAML, AWS, Azure DevOps, VS Code, IntelliJ, Logisim
Languages: French (native), German (native), English (fluent)

PROJECTS

- Deep RL for Settlers of Catan** | *Python, PyTorch, Gym, PPO, DQN* Jan 2025 – Apr 2025
- Used Python Settlers of Catan simulator with Gym wrappers to support deep RL agents
 - Trained PPO and DQN agents with custom architectures on the game's partially observable, large action space
 - Designed league-style self-play evaluation, achieving a 45% win rate with PPO agents over recent versions
 - Compared PPO vs. DQN, highlighting PPO stability and trading/long-horizon challenges
- Deep RL for Chess (AlphaZero-inspired)** | *Python, PyTorch, MCTS, CNN* May 2025 – Sep 2025
- Implemented self-play RL chess AI (AlphaZero) with residual CNN predicting policy and value
 - Integrated Monte Carlo Tree Search (MCTS) guided by the network for strategic move selection
 - GPU-accelerated training and parallel self-play for scalable learning
 - Built interactive human vs. AI interface; tracked performance improvements over iterations
- Media Coverage Analysis Pipeline** | *Python, NLP, TF-IDF, APIs* Nov 2025 – Dec 2025
- Built a modular Python pipeline to collect, clean, and analyze large-scale North American news coverage of political figures using TheNewsAPI
 - Designed standardized TSV workflows to support qualitative coding, sentiment annotation, and dataset merging for mixed-methods analysis
 - Implemented global unigram TF-IDF topic characterization to identify salient themes across manually coded media categories
 - Developed automated visualization and reporting tools to quantify sentiment, topic distributions, and media source biases
- Robe&Hood – Indie Video Game** | *C#, Unity, A** May 2023 – Present
- Designed and developed a 2D stealth-action game in Unity, implementing core gameplay systems including player movement, combat, enemy AI, dynamic scene-switching and game management
 - Implemented pathfinding and decision-making for NPCs using A* path-finding and FSMs for state-based AI, enabling adaptive enemy behavior
 - Engineered reusable C# gameplay systems within Unity, enabling efficient iteration and feature expansion
 - Integrated level design, animation, and gameplay feedback loops to improve player experience and difficulty balancing

VOLUNTEERING / EXTRACURRICULARS

- Chess Club VP Internal** Sep 2023 – Present
McGill Student's Chess Club Montreal, QC
- Organized and ran weekly meetings and chess tournaments; coordinated logistics for club events and ensured smooth internal operations
 - Served on the Executive Committee, contributing to constitutional decisions and funding applications; supported major events including the Canadian University Chess Championship (Montreal, Jan 2025) and promoted the club at university-wide events such as the McGill Club Fair
- TechFair Volunteer** Oct 2022, Feb 2024, Oct 2024
McGill TechFair Montreal, QC
- Volunteered three times with a 50+ student team to set up the event space, assist company representatives, and ensure smooth attendee flow throughout the day
- GameDev McGill Member** Jan 2024 – Present
GameDev McGill Montreal, QC
- Won 3rd place of the 2024 edition of the student-run 48h game development hackathon 'McGameJam' with our game 'Rube's Rocket Rumble'
 - Approved members of the club attend bi-weekly meetings where members show off their games, talks led by industry experts from companies like Ubisoft and EA Sports, and coding bootcamps